

# IMPERIAL



## Recent changes in the Firedrake software ecosystem

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16/09/2025

- Firedrake '22** pyop3: A new domain-specific language for automating high-performance mesh-based simulation codes
- Firedrake '23** Latest developments in pyop3
- Firedrake '24** pyop3 is coming
- Firedrake USA '25** pyop3: a better code generator for mesh-based computations



# Submit thesis/Begin postdoc

10/24



# New Mac runners

11/24



# Absorb PyOP2 and TSFC in the Firedrake repository

12/24



# pip install firedrake

02/25



# How did we get here?

- Firedrake needs a particular environment of compiled dependencies
- Most users are unfamiliar with compiling complex software
- HPC makes this even harder
- Hence `firedrake-install`



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- `firedrake-install` was not a robust solution
- Install failures were extremely hard to debug



# What happened?

- `pip install libsupermesh`
- 3 stage install with `firedrake-configure` (intentionally stupid)
- OpenMPI instead of MPICH (`firedrake-run-split-tests`)
- `firedrake-check`
- `--netgen` etc into `pyproject.toml`
- New Docker images



# Success stories

- Much easier HPC install
- PETSc `--download-firedrake`
- NixOS package



# New Linux runners

02/25

2 × ...

- 384GB RAM
- NVIDIA A100 GPU
- 4 runners



# Smoke tests

02/25

- Gusto
- Thetis
- G-ADOPT
- Spyro

It really works!



# How to add smoke tests for your library

```
- name: Run Gusto smoke tests
  # Only test Gusto in real mode
  if: |
    inputs.run_tests &&
    (success() || steps.install.conclusion == 'success') &&
    matrix.arch == 'default'
  run: |
    . venv/bin/activate
    git clone --depth 1 https://github.com/firedrakeproject/gusto.git gusto-repo
    pip install --verbose ./gusto-repo
    python -m pytest -n 8 --verbose \
      gusto-repo/integration-tests/balance/test_saturated_balance.py \
      gusto-repo/integration-tests/equations/test_thermal_sw.py \
      gusto-repo/integration-tests/transport/test_embedded_dg_advection.py
    timeout-minutes: 10
```



02/25



# mpi-pytest

```
def test_comm_world_size_equals_two():  
    assert COMM_WORLD.size == 2
```

```
def test_comm_world_size_equals_three():  
    assert COMM_WORLD.size == 3
```

```
$ pytest test_comms.py # won't work!
```

```
$ mpiexec -n 2 pytest test_comms.py # won't work!
```





# mpi-pytest

```
@pytest.mark.parallel(2)
def test_comm_world_size_equals_two():
    assert COMM_WORLD.size == 2
```

\$ pytest test\_comms.py *# works, calls MPI under the hood*

\$ mpiexec -n 2 pytest test\_comms.py -m parallel[2] *# works*

```
@pytest.mark.parallel(3)
def test_comm_world_size_equals_three():
    assert COMM_WORLD.size == 3
```



# mpi-pytest

```
@pytest.mark.parallel(2)
def test_comm_world_size_equals_two():
    assert COMM_WORLD.size == 2

@pytest.mark.parallel(3)
def test_comm_world_size_equals_three():
    assert COMM_WORLD.size == 3
```

\$ pytest test\_comms.py *# works, calls MPI under the hood*

\$ mpiexec -n 2 pytest test\_comms.py -m parallel[2] *# works*

pip install mpi-pytest



# Stop using our own fork of PETSc

03/25



# First Firedrake release

04/25



# Why it happened

- To provide more stability for downstream packages



## How it happened

1. Coordinated the release of UFL, pyadjoint, loopy
2. Made the first releases of FIAT and libsupermesh
3. 2025.4.0 released on PyPI



## How it happened

1. Coordinated the release of UFL, pyadjoint, loopy
2. Made the first releases of FIAT and libsupermesh
3. 2025.4.0 released on PyPI
4. It immediately broke
5. 2025.4.0.post0 released



# Understanding Firedrake's release strategy





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# Understanding Firedrake's release strategy

2025.4.0      release

main



# Understanding Firedrake's release strategy

|          |          |                   |
|----------|----------|-------------------|
|          | docs     | breaking change 1 |
|          | bugfix 1 | new feature 1     |
| 2025.4.0 | release  | main              |

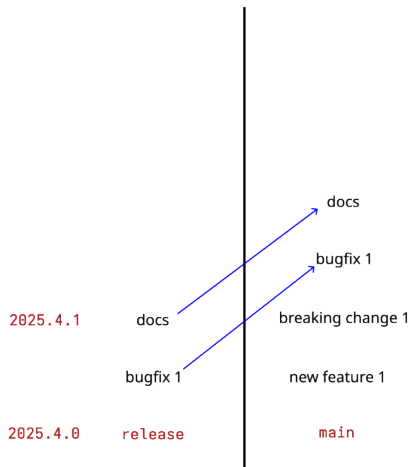


# Understanding Firedrake's release strategy

|          |          |                   |
|----------|----------|-------------------|
| 2025.4.1 | docs     | breaking change 1 |
|          | bugfix 1 | new feature 1     |
| 2025.4.0 | release  | main              |



# Understanding Firedrake's release strategy



# Understanding Firedrake's release strategy

2025.4.1

docs

bugfix 1

2025.4.0

release

docs

bugfix 1

breaking change 1

new feature 1

main



# Understanding Firedrake's release strategy

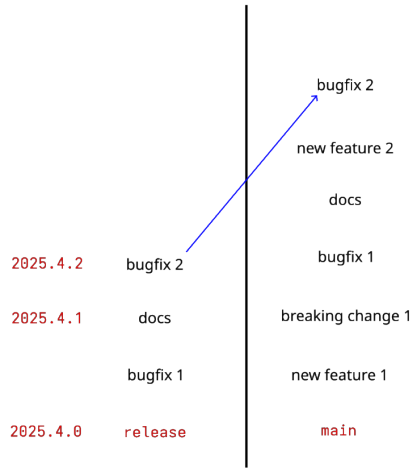


# Understanding Firedrake's release strategy





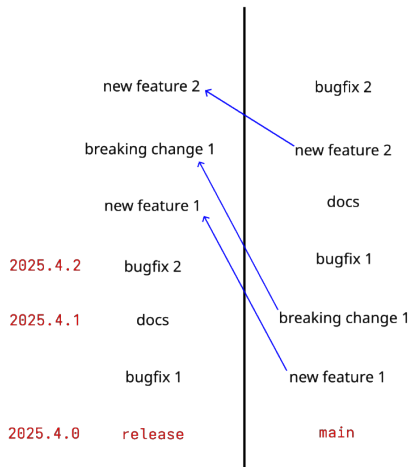
# Understanding Firedrake's release strategy



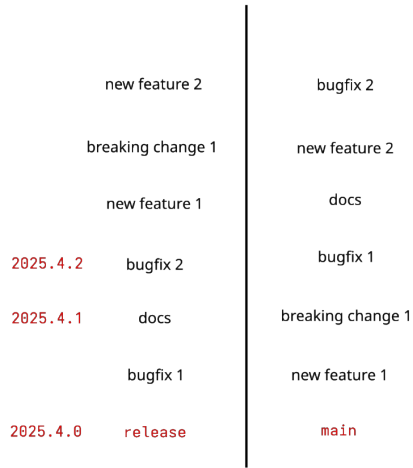
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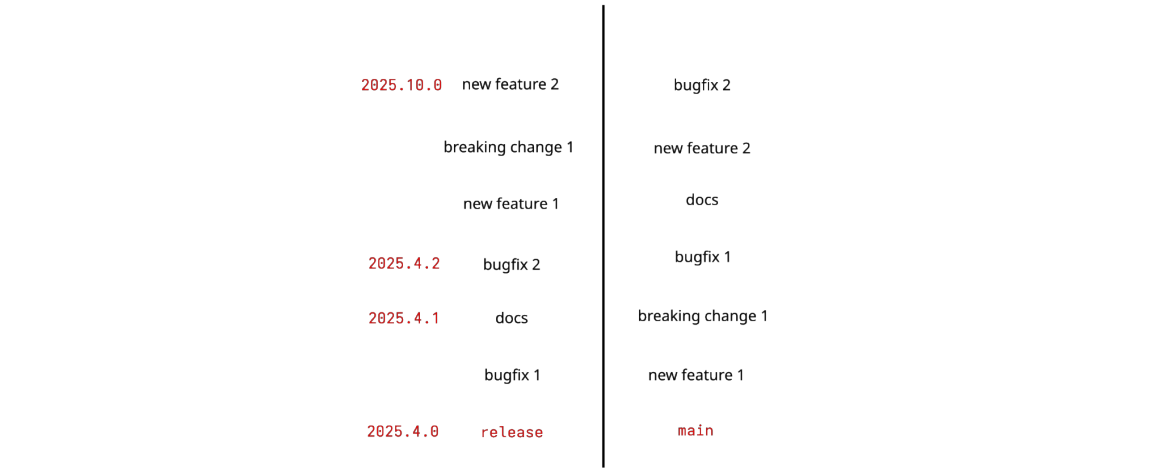
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# Supporting main and/or release



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Strategy 1 Just pick one.



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Strategy 2 Maintain your own `release` and `main` branches.

Strategy 3 Check the version.

```
if importlib.metadata.version("firedrake") > "2025.4":  
    # new API  
else:  
    # old API
```



## Supporting `main` and/or release

Strategy 1 Just pick one.

Strategy 2 Maintain your own `release` and `main` branches.

Strategy 3 Check the version.

```
if importlib.metadata.version("firedrake") > "2025.4":  
    # new API  
else:  
    # old API
```

We have Docker images for `release` (`firedrake-vanilla-default:dev-release`) and `main` (`firedrake-vanilla-default:dev-main`) that you can use for your CI.



# Use more system packages when building PETSc

04/25

From:

Depending on your platform, PETSc may take an hour or more to build!

To:

- |   |                                   |        |
|---|-----------------------------------|--------|
| > | ✓ Install system dependencies (2) | 37s    |
| > | ✓ Install PETSc                   | 2m 21s |
| > | ✓ Install Firedrake               | 9m 0s  |



07/25



# Motivation

The screenshot shows a GitHub pull request interface. At the top, the repository is 'pyadjoint' by 'dolphin-adjoint'. The pull request is titled 'Use petsctools #214' and is in the 'Open' state. It shows that 'jrmaddison' wants to merge 1 commit into 'dolphin-adjoint:master' from 'jrmaddison:jrmaddison/petsctools'. The pull request has 1 conversation, 1 commit, 0 checks, and 1 file changed. A comment by 'jrmaddison' is visible, stating 'No description provided.' The pull request is being reviewed by 'connorjward'. The code diff shows a modification to the 'flatten\_parameters' function in 'firedrake/petsc.py'.

Use petsctools #214

[Open](#) jrmaddison wants to merge 1 commit into `dolphin-adjoint:master` from `jrmaddison:jrmaddison/petsctools`

Conversation 1 Commits 1 Checks 0 Files changed 1 +7 -200

jrmaddison commented yesterday

Member

No description provided.

1

Reviewers

connorjward

Still in progress? [Convert to draft](#)

```
130 - # Modified version of flatten_parameters function from firedrake/petsc.py,
131 - # Firedrake master branch 57e21cc8ebdb044c1d8423b48f3dbf70975d5548, first
132 - # added 2024-08-08
133 - def flatten_parameters(parameters, sep="_"):
134 -     """Flatten a nested parameters dict, joining keys with sep.
```



# Features

- Interacting with PETSc options: `OptionsManager`
- Passing arbitrary data through PETSc to Python preconditioners: 'app context'
- Handling citations
- Help initialising PETSc
- Reading PETSc configuration information

`pip install petsctools`

<https://firedrakeproject.org/petsctools>



## Final remarks

- A lot has changed in Firedrake in the past year
- Feedback welcome on what the community wants next (`pip install petsc?`, `conda?`, `spack?`)

